Roadmap. The last lecture. §58–59 (pp. 1–4): photon rules added to the Dirac rules (LecNote19–20); $e^+e^- \to \gamma\gamma$, one trace left (result noted on p. 3). §62 (pp. 4–9): counterterms, vacuum polarization in full, on-shell Z_3; Z_2, Z_m, Z_1 quoted. §66 (p. 10): β via $Z_1 = Z_2$ (Ward identity, §67–68, not covered). Supplements: GammaTraces ($e^+e^- \to \gamma\gamma$ to σ), RunningCoupling (QED's $\alpha(\mu)$), PhotonGauges, DiracFeynmanRules, DimRegMSbar, Renormalizability.

Lec 23

Having understood QED, the Feynman Rules should be straightforward - just add the following to before.

Add, Each photon line (wavy lines)

Internal photons, we have

$$\frac{-i\not{p}\,g^{\mu\nu}}{k^2 - i\varepsilon}$$

[here $p^{\mu\nu} = g^{\mu\nu}$, typically most convenient. Feynman Gauge].

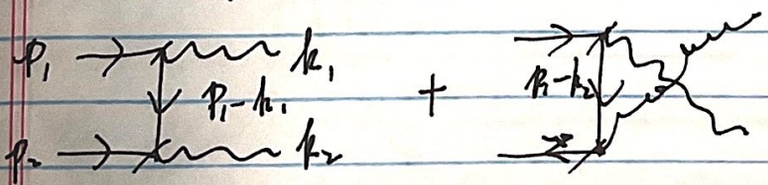
Incoming photons $\varepsilon^{\mu}_{\lambda_i}{}^{*}(\vec{k_i})$

Outgoing photons $\varepsilon^{\mu}_{\lambda_i}(\vec{k_i})$.

And each ~~dot~~ vertex is $ie\gamma^\mu$.

Note that typically one need to pay attention to a Fermion "order" as there are hidden indecies ~~it~~ that we didn't spell out. Photons can be moved around.

Let's calculate a simple process. $e^+e^- \to \gamma\gamma$.
(A pair of particle antiparticle annihilate into two photons)



$$i\mathcal{T} = \bar{v}_{s_2}(p_2)\, ie\gamma_\nu \frac{-i(\not{p}_1 + \not{k}_1 + m)}{(p_1 - k_1)^2 + m^2 - i\varepsilon}\, ie\gamma_\mu\, u_{s_1}(p_1)\, \varepsilon^{*\mu}_{\lambda_1}(k_1)\, \varepsilon^{\nu}_{\lambda_2}(k_2)$$

$$+ \cdots$$

$$\mathcal{T} = e^2 \varepsilon_1^\mu \varepsilon_2^\nu \bar{v}_{s_2} \left[\gamma_\nu \frac{-\not{p}_1 + \not{k}_1 + m}{(p_1 - k_1)^2 + m^2 - i\varepsilon} \gamma_\mu + \gamma_\mu \frac{-\not{p}_1 + \not{k}_2 + m}{(p_1 - k_2)^2 + m^2 - i\varepsilon} \gamma_\nu \right] u_{s_1}$$

as $t = -(p_1 - k_1)^2$

$u = -(p_1 - k_2)^2$

$$\mathcal{T} = e^2 \varepsilon_1^\mu \varepsilon_2^\nu \bar{v}_{s_2} \left[\frac{-\not{p}_1 + \not{k}_1 + m}{(m^2 - t)} + \frac{-\not{p}_1 + \not{k}_2 + m}{(m^2 - u)} \right] u_{s_1}$$

$|\mathcal{T}|^2 =$ four terms.

$$e^4 \varepsilon_1^\mu \varepsilon_1^{*\mu'} \varepsilon_2^\nu \varepsilon_2^{*\nu'}$$

$$\times \left\{ \frac{\#\not{k}\not{k} + t}{(t - m^2)^2} + \frac{\#_{uu}}{(u - m^2)^2} + \frac{\#_{tu} + \#_{ut}}{(t - m^2)(u - m^2)} \right\}$$

Let we demonstrate one, say, as usual, we are interested in the spin & polarization ~~averaged~~ averaged summed cross section.

$$\frac{1}{4} \overline{\sum_{s_1, s_2 = \pm}} \; \overline{\sum_{\lambda_1, \lambda_2 = \pm}} \; |\mathcal{T}|^2$$

for the $\#\not{k}\not{k} + t$.

we have

$$\frac{1}{4}\, e^4 \cdot g^{\mu\mu'} g^{\nu\nu'} \bar{v}_2 \gamma_\nu \frac{-\not{p}_1 + \not{k}_1 + m}{-(t - m^2)} \gamma_\mu u_1 \quad (1)$$

$$\times \left(\bar{v}_2 \gamma_{\nu'} \frac{-\not{p}_1 + \not{k}_1 + m}{-(t - m^2)} \gamma_{\mu'} u_1\right)^*$$

~~$= e^4 \bar{v}_2 \gamma_\nu \gamma_{\nu'}$~~

$$= \frac{1}{4} \sum_{s_1, s_2} \frac{e^4}{(t - m^2)^2} \cdot \bar{v}_2 \gamma_\nu (-\not{p}_1 + \not{k}_1 + m) \gamma_\mu u_1 \bar{u}_1 \gamma^\mu (-\not{p}_1 + \not{k}_1 + m) \gamma^\nu v_2$$

$$= \frac{1}{4} \sum_{s_1, s_2} \frac{e^4}{(t - m^2)^2} \mathrm{Tr}\left[(-\not{p}_2 - m) \gamma_\nu (-\not{p}_1 + \not{k}_1 + m) \gamma_\mu (-\not{p}_1 + m) \gamma^\mu (-\not{p}_1 + \not{k}_1 + m) \gamma^\nu\right]$$

using $\gamma_\mu \not{a} \gamma^\mu = +2\not{a}$

$\gamma_\mu \gamma^\mu = -4.$

$$= \frac{e^4}{4(t - m^2)^2} \mathrm{Tr}\left[(-2\not{p}_2 + 4m)(-\not{p}_1 + \not{k}_1 + m)(-2\not{p}_1 - 4m)(-\not{p}_1 + \not{k}_1 + m)\right]$$

$$= \frac{e^4}{4(t - m^2)^2} \left[\mathrm{Tr}[4\gamma_s] + m^2 \mathrm{Tr}[2\gamma_s] + m^4 \cdot (-64)\right] \quad (2)$$

$\geq$ (Also notice that $\not{p}_1 \not{p}_1 = m^2$. one can reduce the # of 4 γ_s trace that needs to be done!

1 Why $g^{\mu\mu'}$: for $\mathbf{k} \parallel \hat{z}$, $\sum_\lambda \varepsilon^\mu_\lambda \varepsilon^{\nu *}_\lambda = \mathrm{diag}(0,1,1,0) = g^{\mu\nu} - (k^\mu \bar{k}^\nu + \bar{k}^\mu k^\nu)/k \cdot \bar{k}$, $\bar{k} = (\omega, -\mathbf{k})$. The extra terms carry k^μ or k^ν and vanish since $k_1^\mu \mathcal{T}_{\mu\nu} = 0$ (Ward identity), true for the sum of both diagrams, not for each: use g only once all four terms are kept.

2 Finished (all four terms, polarization sums $\to g$):

$$\langle |\mathcal{T}|^2 \rangle = 2e^4 \left[\frac{m^2 - u}{m^2 - t} + \frac{m^2 - t}{m^2 - u} + 4m^2 c - 4m^4 c^2\right],$$

$c = \frac{1}{m^2 - t} + \frac{1}{m^2 - u}$; symmetric under $t \leftrightarrow u$, and $2e^4(u/t + t/u)$ for $m \to 0$. Checked against explicit spinors and physical polarizations.

eventually, we assemble everything and make it an observable cross section.

$$d\sigma = \frac{1}{\text{flux}} \cdot |T|^2 \cdot d\text{PS}_2 .$$

$d\text{PS}_2$: two photons are identical particles, phase space have $1/2$ factor in it.

Now, let's work out the loop corrections.

We begin with

~~$\mathcal{L}_1 = \mathcal{L}_2$~~

~~$\mathcal{L}_0 = i\bar{\psi}\partial\!\!\!/\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - m\bar{\psi}\psi$~~

$$\mathcal{L}_0 = i\bar{\psi}\partial\!\!\!/\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - m\bar{\psi}\psi$$

$$\mathcal{L}_1 = Z_1 e \bar{\psi} A\!\!\!/ \psi + \mathcal{L}_{CT}$$

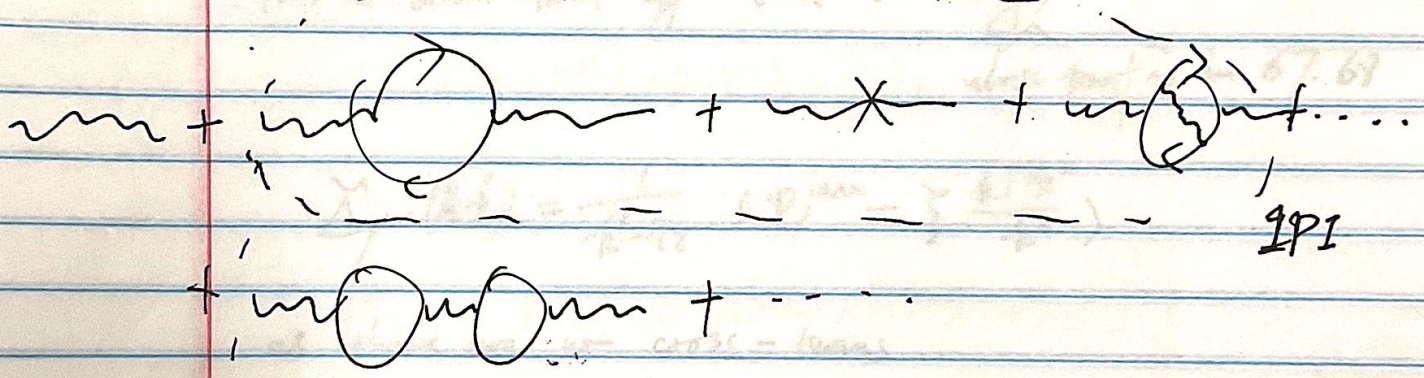
$$\mathcal{L}_{CT} = i(Z_2 - 1)\bar{\psi}\partial\!\!\!/\psi - (Z_m - 1)\bar{\psi}\psi m$$

$$- \frac{1}{4}(Z_3 - 1)F^{\mu\nu}F_{\mu\nu}.$$

1 Which ones: power counting, $\omega = 4 - E_A - \frac{3}{2}E_\Psi$, also allows 1-, 3- and 4-photon divergences (Renormalizability). Odd photon numbers vanish: each fermion loop cancels its reversed orientation (Furry's theorem, C invariance). Gauge invariance forbids a photon mass ($\Pi^{\mu\nu}$ transverse, p. 6) and makes the 4-photon divergence cancel in the sum over orderings. So Z_1, Z_2, Z_m, Z_3 (p. 4) suffice.

This is a renormalizable theory with finite # of counter term needed.
Other counter terms do not exist by various reasons (symmetries, etc).

Let's do one calculation for example.
What is the exact propagator for the photons?
(also known as "vacuum polarization" & "self energy" correction")



the exact propagator (in momentum space)

$$\tilde{\Delta}_{\mu\nu}(k) = \tilde{\Delta}_{\mu\nu}(k) + \tilde{\Delta}_{\mu\rho}(\vec{k})\,\Pi^{\rho\sigma}(\vec{k})\,\tilde{\Delta}_{\sigma\nu}(\vec{k}) + \cdots$$

In the general R_ξ gauge

$$\tilde{\Delta}_{\mu\nu}(k) = \frac{1}{k^2 - i\varepsilon}\left(\delta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2}\right).$$

Physical result shall not depend on ξ, implying the result should be "transverse", e.g.

$$k_\mu \Pi^{\mu\nu} = k_\nu \Pi^{\mu\nu} = 0.$$

such that

$$\Pi^{\mu\nu}(k) = \Pi(k)(k^2 g^{\mu\nu} - k^\mu k^\nu)$$
$$= k^2 \Pi(k) P^{\mu\nu}.$$

(this is Guaranteed by Ward Identity) whose proof is in 67.68

$$\tilde{\Delta}_{\mu\nu}(k) = \frac{1}{k^2 - i\xi}\left(P^{\mu\nu} - \xi \frac{k^\mu k^\nu}{k^2}\right) \quad (1)$$

as there are no cross-terms

$$\tilde{\Delta}_{\mu\nu}(k) = \frac{P_{\mu\nu}(k)}{k^2[1-\Pi(k)] - i\xi} + \xi \frac{k_\mu k_\nu / k^2}{k^2 - i\xi}$$

Again the second term should be physically irrelevant, we can drop it or set $\xi = 0$ (Lorentz Gauge)

For on-shell renormalization scheme, we should have

$$\Pi(0) = 0 \qquad \boxed{\Pi'(k^2)\big|_{k^2=0} = 0} \quad (2)$$

1 Should read $(P^{\mu\nu} + \xi k^\mu k^\nu/k^2)/(k^2 - i\epsilon)$, with $P^{\mu\nu} = g^{\mu\nu} - k^\mu k^\nu/k^2$ here (p. 1's P was g); the lines below have $+\xi$. Since $P^{\mu\nu}k_\nu = 0$, Π never touches the ξ term: "no cross-terms".

2 Only $\Pi(0) = 0$ is a condition: it fixes the one counterterm Z_3 (residue 1); the pole stays at $k^2 = 0$ by transversality. Drop the boxed one: p. 9's result has $\Pi'(0) = e^2/60\pi^2 m^2 \neq 0$.

Let's calculate.

$$i\Pi^{\mu\nu}(k) = \underbrace{(-1)}_{\text{fermion loop}} \underbrace{(ie)^2}_{\text{vertices}} \underbrace{\left(\frac{1}{i}\right)^2}_{\text{propagators}} \int \frac{d^4\ell}{(2\pi)^4}$$

$$\mathrm{Tr}[\tilde{S}(\ell+k)\not{\varepsilon}\gamma^\mu \tilde{S}(\ell)\gamma^\nu]$$

$$-i(Z_3-1)(k^2 g^{\mu\nu} - k^\mu k^\nu) + O(e^4).$$

Now

$$\mathrm{Tr}[\tilde{S}(\ell+k)\gamma^\mu \tilde{S}(\ell)\gamma^\nu] = \int_0^1 dx \frac{4N^{\mu\nu}}{(q^2+D)^2}$$

$$D = x(1-x)k^2 + m^2 - i\varepsilon$$

$$4N^{\mu\nu} = \mathrm{Tr}[(-\not{\ell} - \not{k} + m)\gamma^\mu(-\not{\ell}+m)\gamma^\nu]$$

$$= 4[(\ell+k)^\mu \ell^\nu - (\ell+k)\cdot\ell\, g^{\mu\nu} - (\ell+k)^\nu \ell^\mu + \not{\ell} - m^2 g^{\mu\nu}]$$

[1] Should read $+(\ell+k)^\nu \ell^\mu$: $\mathrm{Tr}[\not{a}\gamma^\mu \not{b}\gamma^\nu] = 4(a^\mu b^\nu + a^\nu b^\mu - a\cdot b\, g^{\mu\nu})$, symmetric in $\mu\nu$. The shifted line below $(2q^\mu q^\nu)$ uses $+$.

Recall that we shift the momentum $\ell \to \cancel{\ell + xk}$

$$q = \ell + xk.$$

$$N^{\mu\nu} \to 2q^\mu q^\nu - 2x(1-x)k^\mu k^\nu - [q^2 - x(1-x)k^2 + m^2]g^{\mu\nu}$$

This integral diverges, we need to "regulate" it.

We can use dim-reg. $d = 4 - \varepsilon$

("Recall" 14.5) $\int d^d q\, q^\mu q^\nu f(q^2) = \frac{1}{d} g^{\mu\nu} \int d^d q\, q^2 f(q^2)$

$$\int d^d q\, q^\mu f(q^2) = 0$$

. . .

$$N^{\mu\nu} \to -2x(1-x) k^\mu k^\nu + [\left(\tfrac{2}{d} - 1\right) q^2 + x(1-x) k^2 + A - m^2] g^{\mu\nu}$$

"before" we do the calculation, let's observe that.

$$\left(\frac{2}{d} - 1\right) \int \frac{d^d q}{(2\pi)^4} \frac{q^2}{(q^2 + D)^2} = D \int \frac{d^d q}{(2\pi)^4} \frac{1}{(q^2 + D)^2}$$ [1]

by comparing the "master equation"

It is then clear that

$$N^{\mu\nu} \to 2x(1-x)\left(k^2 g^{\mu\nu} - k^\mu k^\nu\right)$$

telling us that $\Pi^{\mu\nu}(k)$ is transverse!

Now we can evaluate.

$$\mu^{4-\varepsilon} \int_0^1 \frac{d^d q}{(2\pi)^d} \frac{1}{(q^2 + D)^2} = \frac{i}{16\pi^2} \Gamma(\varepsilon/2) \left(\frac{4\pi\mu^2}{D}\right)^{\varepsilon/2}$$

$$= \frac{i}{8\pi^2} \left[\frac{1}{\varepsilon} - \frac{1}{2} \ln(D/\mu^2)\right] + \ldots$$

$$\mu^2 = 4\pi e^{-\gamma} \mu^2$$

[1] A dim-reg identity: the master formula gives $\int q^2/(q^2+D)^2 = [d/(2-d)]\, D \int 1/(q^2+D)^2$ for any d. With a 4-d cutoff the two sides differ by a term $\propto \Lambda^2$, which would leave $\Lambda^2 g^{\mu\nu}$ in $\Pi^{\mu\nu}$: a photon mass, breaking the Ward identity. Dim reg drops pure powers of Λ and keeps $\Pi^{\mu\nu}$ transverse (DimRegMSbar, Renormalizability).

So we know that

$$\Pi(k^2) = -\frac{e^2}{\pi^2}\int_0^1 dx\,(1-x)\times\left[\frac{1}{\varepsilon} - \frac{1}{2}\ln(D/\mu^2)\right] - (Z_3 - 1) + O(e^4)$$

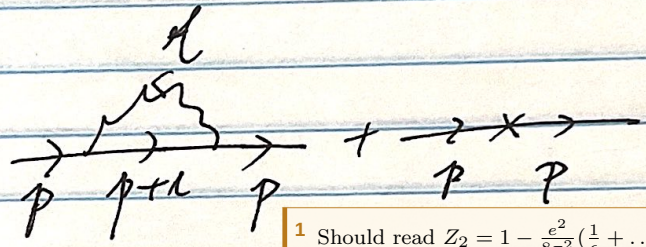
Imposing on-shell condition yields that

$$Z_3 = 1 - \frac{e^2}{6\pi^2}\left[\frac{1}{\varepsilon} - \ln(m/\mu)\right] + O(e^4)$$

&

$$\Pi(k^2) = \frac{e^2}{2\pi^2}\int_0^1 dx\, x(1-x)\ln(D/m^2) + O(e^4).$$

Similarly, we can do



can get Z_2, Z_m.

1 Should read $Z_2 = 1 - \frac{e^2}{8\pi^2}(\frac{1}{\epsilon} + \dots)$ (Feynman gauge, like Z_m). The last line is the vertex factor $Z_1 = 1 - \frac{e^2}{8\pi^2}(\frac{1}{\epsilon} + \dots) + O(e^4)$, not Z_3 (fixed above): $Z_1 = Z_2$ (Ward identity), as p. 10 needs.

~~In~~ there are some IR subtitlies here but can be circumvented by giving γ a small mass and take $m_\gamma \to 0$ in the last step.

We can get

$$Z_2 = 1 - \frac{e^2}{6\pi^2}\left(\frac{1}{\varepsilon} + \text{finite}\right) + O(e^4)$$ (1)

$$Z_m = 1 - \frac{e^2}{2\pi^2}\left(\frac{1}{\varepsilon} + \text{finite}\right) + O(e^4).$$

and similarly

$$Z_3 = 1 - \frac{e^2}{2\pi^2}\left(\frac{1}{\varepsilon} + \text{finite}\right) + O(\varepsilon^4).$$

We can also work out the beta function

$$e_0 = Z_3^{-1/2} Z_2^{-1} Z_1 \mu^{\epsilon/2} e.$$

$$\alpha_0 = Z_3^{-1} Z_2^{-2} Z_1^2 \mu^{\epsilon} \alpha.$$

In $\overline{\text{MS}}$ one can see that

$$\beta(\alpha) = \frac{2\alpha^2}{3\pi} + O(\alpha^3). \text{ (1)}$$

1 Where $2\alpha^2/3\pi$ comes from: $Z_1 = Z_2$ (Ward identity; p. 9's last line is Z_1), so $\alpha_0 = Z_3^{-1}\tilde{\mu}^{\epsilon}\alpha$, with MS-bar $Z_3 = 1 - 2\alpha/3\pi\epsilon$ (p. 9's pole). $d\alpha_0/d\ln\mu = 0$ gives $d\alpha/d\ln\mu = -\epsilon\alpha + 2\alpha^2/3\pi$. Same rate as p. 9's $\Pi(k^2) \to (\alpha/3\pi)\ln(k^2/m^2)$ at $k^2 \gg m^2$: the charge $\alpha/(1-\Pi)$ grows at short distance.